DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

# WO-200-02 — Internal inspection of vertical vessel V-101

| | | | |
|---|---|---|---|
| **Area** | A200 Vessels and column | **Task** | VI Internal inspection |
| **Priority** | 2 (integrity) | **Issued by** | Facility owner (demonstration) |
| **Equipment** | V-101 | **Lines** | 4"-P300-P-1005, 6"-P300-P-1006, 1"-P150-V-6001, 1"-P150-D-6002, 1"-P150-N-3011, 3/4"-P300-I-6003, 3/4"-P300-I-6004, 3/4"-P300-SP-6005 |
| **Isometrics** | ISO-4"-P300-P-1005, ISO-6"-P300-P-1006, ISO-1"-P150-V-6001, ISO-1"-P150-D-6002, ISO-1"-P150-N-3011, ISO-3/4"-P300-I-6003, ISO-3/4"-P300-I-6004, ISO-3/4"-P300-SP-6005 | | |
| **Permits** | General work permit, Confined-space entry permit | | |

## Reason

Scheduled internal inspection of the vessel.

## Scope of work

1. Isolate V-101 per the isolation list; drain, vent, purge and gas-test for entry.
2. Open for entry; clean as needed for inspection.
3. Internal visual examination; ultrasonic thickness at the 14 TMLs.
4. Close with new gaskets on every opened joint; remove blinds; restore the line-up.

Affected items: V-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6

## Isolation list (SW-WO-3)

| # | Kind | Tag | Line | Action | Location | Fitted ✓ | Removed ✓ |
|---|---|---|---|---|---|---|---|
| 1 | valve | 4"-P300-P-1005/B2 | 4"-P300-P-1005 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1005/B2) on V-101-N1 | E 33'-6" N -20'-8-3/16" EL 121'-0" | | |
| 2 | blind | V-101-N1 | 4"-P300-P-1005 | Fit a blind between 4"-P300-P-1005/B2 and the nozzle flange V-101-N1 | E 33'-6" N -21'-3-15/16" EL 121'-0" | | |
| 3 | valve | 6"-P300-P-1006/A2 | 6"-P300-P-1006 | Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1006/A2) on V-101-N2 | E 33'-6" N -20'-8-9/16" EL 109'-0" | | |
| 4 | blind | V-101-N2 | 6"-P300-P-1006 | Fit a blind between 6"-P300-P-1006/A2 and the nozzle flange V-101-N2 | E 33'-6" N -21'-3-11/16" EL 109'-0" | | |
| 5 | valve | 1"-P150-V-6001/2 | 1"-P150-V-6001 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6001/2) on V-101-N3 | E 33'-6" N -25'-0" EL 128'-8-3/4" | | |
| 6 | blind | V-101-N3 | 1"-P150-V-6001 | Fit a blind between 1"-P150-V-6001/2 and the nozzle flange V-101-N3 | E 33'-6" N -25'-0" EL 128'-1-15/16" | | |
| 7 | valve | 1"-P150-D-6002/2 | 1"-P150-D-6002 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6002/2) on V-101-N4 | E 37'-6-1/2" N -25'-0" EL 108'-6" | | |
| 8 | blind | V-101-N4 | 1"-P150-D-6002 | Fit a blind between 1"-P150-D-6002/2 and the nozzle flange V-101-N4 | E 36'-11-11/16" N -25'-0" EL 108'-6" | | |
| 9 | valve | 1"-P150-N-3011/B2 | 1"-P150-N-3011 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-N-3011/B2) on V-101-N5 | E 37'-6-1/2" N -25'-0" EL 120'-6" | | |
| 10 | blind | V-101-N5 | 1"-P150-N-3011 | Fit a blind between 1"-P150-N-3011/B2 and the nozzle flange V-101-N5 | E 36'-11-11/16" N -25'-0" EL 120'-6" | | |
| 11 | valve | 3/4"-P300-I-6003/2 | 3/4"-P300-I-6003 | Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6003/2) on V-101-N6 | E 37'-6-1/4" N -25'-0" EL 112'-6" | | |
| 12 | blind | V-101-N6 | 3/4"-P300-I-6003 | Fit a blind between 3/4"-P300-I-6003/2 and the nozzle flange V-101-N6 | E 36'-11-9/16" N -25'-0" EL 112'-6" | | |
| 13 | valve | 3/4"-P300-I-6004/2 | 3/4"-P300-I-6004 | Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6004/2) on V-101-N7 | E 37'-6-1/4" N -25'-0" EL 118'-0" | | |
| 14 | blind | V-101-N7 | 3/4"-P300-I-6004 | Fit a blind between 3/4"-P300-I-6004/2 and the nozzle flange V-101-N7 | E 36'-11-9/16" N -25'-0" EL 118'-0" | | |
| 15 | valve | 3/4"-P300-SP-6005/2 | 3/4"-P300-SP-6005 | Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-SP-6005/2) on V-101-N8 | E 37'-6-1/4" N -25'-0" EL 110'-6" | | |
| 16 | blind | V-101-N8 | 3/4"-P300-SP-6005 | Fit a blind between 3/4"-P300-SP-6005/2 and the nozzle flange V-101-N8 | E 36'-11-9/16" N -25'-0" EL 110'-6" | | |

## Inspections

| Id | Inspection | Criteria | Basis |
|---|---|---|---|
| WO-200-02-I1 | Internal visual examination of the shell, heads and nozzles | Findings recorded against the owner's integrity criteria | SW-NDE-4 |
| WO-200-02-I2 | Ultrasonic thickness at the 14 TMLs listed | Each reading above the owner's minimum wall | SW-WO-6 |

## Thickness monitoring locations (SW-WO-6)

| TML | Item | Location | Position | Nominal wall | R1 | R2 | R3 | R4 |
|---|---|---|---|---|---|---|---|---|
| WO-200-02-TML01 | V-101 | Nozzle neck V-101-N1 | E 33'-6" N -21'-3-15/16" EL 121'-0" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-200-02-TML02 | V-101 | Nozzle neck V-101-N2 | E 33'-6" N -21'-3-11/16" EL 109'-0" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-200-02-TML03 | V-101 | Nozzle neck V-101-N3 | E 33'-6" N -25'-0" EL 128'-1-15/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-200-02-TML04 | V-101 | Nozzle neck V-101-N4 | E 36'-11-11/16" N -25'-0" EL 108'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-200-02-TML05 | V-101 | Nozzle neck V-101-N5 | E 36'-11-11/16" N -25'-0" EL 120'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-200-02-TML06 | V-101 | Nozzle neck V-101-N6 | E 36'-11-9/16" N -25'-0" EL 112'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-200-02-TML07 | V-101 | Nozzle neck V-101-N7 | E 36'-11-9/16" N -25'-0" EL 118'-0" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-200-02-TML08 | V-101 | Nozzle neck V-101-N8 | E 36'-11-9/16" N -25'-0" EL 110'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-200-02-TML09 | V-101 | Bottom head centre | E 33'-6" N -25'-0" EL 105'-2" | | centre: ____ | | | |
| WO-200-02-TML10 | V-101 | Top head centre | E 33'-6" N -25'-0" EL 126'-2-1/2" | | centre: ____ | | | |
| WO-200-02-TML11 | V-101 | Shell at 0.000" above the lower tangent | E 33'-6" N -25'-0" EL 106'-8-1/4" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-200-02-TML12 | V-101 | Shell at 6'-0" above the lower tangent | E 33'-6" N -25'-0" EL 112'-8-1/4" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-200-02-TML13 | V-101 | Shell at 12'-0" above the lower tangent | E 33'-6" N -25'-0" EL 118'-8-1/4" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-200-02-TML14 | V-101 | Shell at 18'-0" above the lower tangent | E 33'-6" N -25'-0" EL 124'-8-1/4" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |

## Test — Service leak test

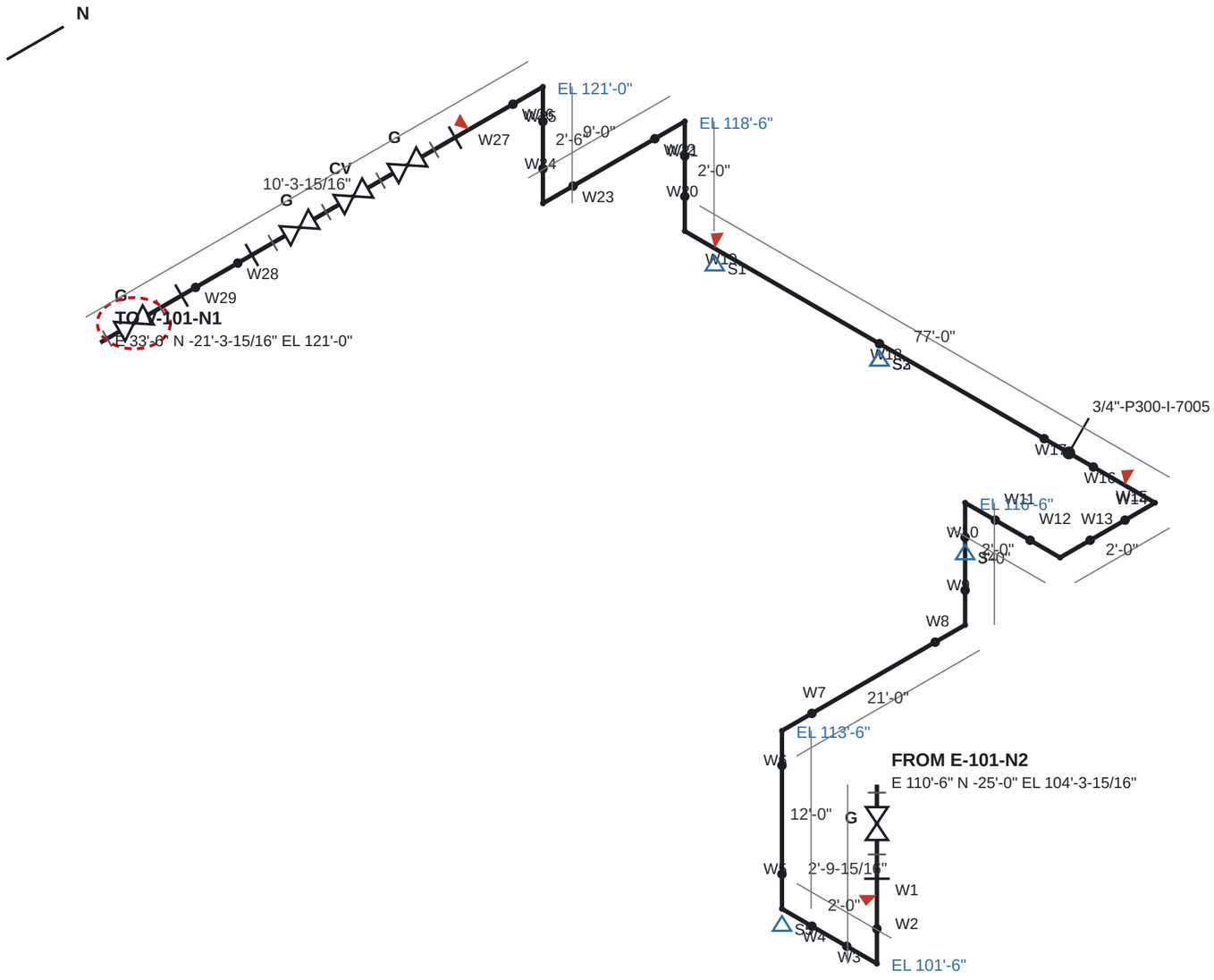
| | |
|---|---|
| **Description** | Every opened joint checked at operating pressure on start-up |
| **Basis** | SW-TEST-2 |

## Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

| # | Activity | Reference | Acceptance | Record | Contr. | Owner | 3rd party | Status | Signed (contr. / owner / 3rd) | Date | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained | SW-WO-3, isolation list | Every isolation in place and tagged; zero pressure at the vent | Signed isolation certificate | H | H | — | | | | |
| 2 | Equipment vented, purged and gas-tested for entry; confined-space entry permit issued | Owner entry procedure | Atmosphere safe for entry per owner limits | Gas test record, entry permit | H | H | — | | | | |
| 3 | Opened for entry; cleaned for inspection | Owner procedure | Surfaces clean enough to inspect | Daily report | S | S | — | | | | |
| 4 | Internal visual examination | SW-NDE-4 | Findings recorded | Inspection report with photographs | W | H | W | | | | |
| 5 | Ultrasonic thickness at 14 TMLs | SW-WO-6 | Above the owner's minimum wall | Thickness record per TML | S | R | R | | | | |
| 6 | Owner's decision on findings (repair, monitor or accept) | Owner integrity procedure | Decision recorded | Signed disposition | R | H | R | | | | |
| 7 | Closed with new gaskets on every opened joint; studs to preload | SW-WO-5 | Preload as stated | Bolt-up record | W | W | — | | | | |
| 8 | Service leak test of opened joints on start-up | SW-TEST-2 | No leak | Leak test record | W | W | — | | | | |
| 9 | Blinds removed, isolations returned to normal line-up per the isolation list | SW-WO-3 | Every blind removed and accounted for | Isolation certificate closed | H | H | — | | | | |
| 10 | Records complete; work order handed back to operations | SW-ITP-3 | All ITP activities accepted or not applicable | Signed ITP and completion record | H | H | R | | | | |

ISOMETRIC — 4"-P300-P-1005 — Internal inspection of vertical vessel V-101




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 4" P300 COMPRESSED SHEET | 8 |
| 2 | VALVE, GATE, WEDGE, OS&Y 4" P300 | 4 |
| 3 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300 | 8 |
| 4 | FLANGE WELD-NECK 4" P300 RAISED FACE | 4 |
| 5 | PIPE 4" P300 BEVELLED ENDS | 124'-6-1/2" |
| 6 | ELBOW 90° R = 5/3 OD 4" P300 | 11 |
| 7 | TEE REDUCING 4" × 3/4" P300 | 1 |
| 8 | VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 | 1 |

## WELDS 29 (4 FIELD) · SUPPORTS 5 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15F W16 W17 W18 W19F W20
W21 W22 W23 W24 W25 W26 W27F W28 W29

## 4"-P300-P-1005

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
E-101-N2 → V-101-N1
PIPE 124'-6-1/2" · WEIGHT 1974 lb

**DWG ISO-4"-P300-P-1005 REV 0 WO-200-02**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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## Isometric ISO-6"-P300-P-1006

ISOMETRIC — 6"-P300-P-1006 — Internal inspection of vertical vessel V-101




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 6" P300 COMPRESSED SHEET | 4 |
| 2 | VALVE, GATE, WEDGE, OS&Y 6" P300 | 2 |
| 3 | STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300 | 4 |
| 4 | FLANGE WELD-NECK 6" P300 RAISED FACE | 3 |
| 5 | PIPE 6" P300 BEVELLED ENDS | 175'-11-5/8" |
| 6 | ELBOW 90° R = 5/3 OD 6" P300 | 8 |
| 7 | TEE REDUCING 6" × 3/4" P300 | 2 |
| 8 | TEE EQUAL 6" P300 | 1 |
| 9 | REDUCER CONCENTRIC 6" × 4" P300 | 1 |
| 10 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300 | 1 |
| 11 | FLANGE WELD-NECK 4" P300 RAISED FACE | 1 |
| 12 | GASKET 4" P300 COMPRESSED SHEET | 1 |

**WELDS 29 (4 FIELD) · SUPPORTS 8 · SPOOLS 6**

W1F W2 W3 W4 W5 W6 W7 W8 W9F W10
W11 W12 W13 W14 W15 W16 W17F W18 W19 W20
W21 W22 W23 W24 W25 W26 W27F W28 W29

## 6"-P300-P-1006

CLASS H300 — Hydrocarbon process
SERVICE P · 6" · P300 · DESIGN 300 psi
V-101-N2 → P-102A-S
PIPE 175'-11-5/8" · WEIGHT 3070 lb

**DWG ISO-6"-P300-P-1006 REV 0 WO-200-02**

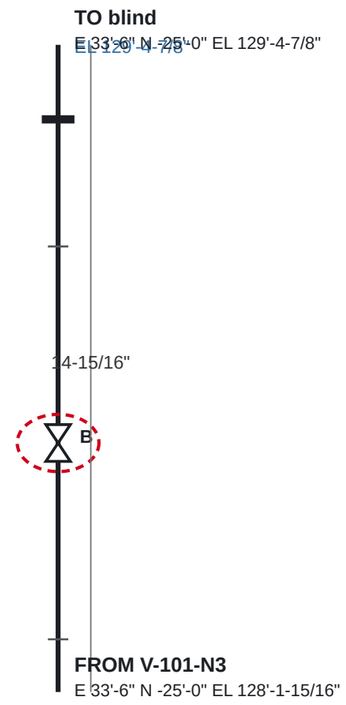
SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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## Isometric ISO-1"-P150-V-6001

ISOMETRIC — 1"-P150-V-6001 — Internal inspection of vertical vessel V-101

N




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 1" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 1" P150 | 1 |
| 3 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 2 |
| 4 | FLANGE BLIND 1" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 1"-P150-V-6001

CLASS H150 — Hydrocarbon process
SERVICE V · 1" · P150 · DESIGN 150 psi
V-101-N3 → blind
PIPE 0.000" · WEIGHT 39 lb

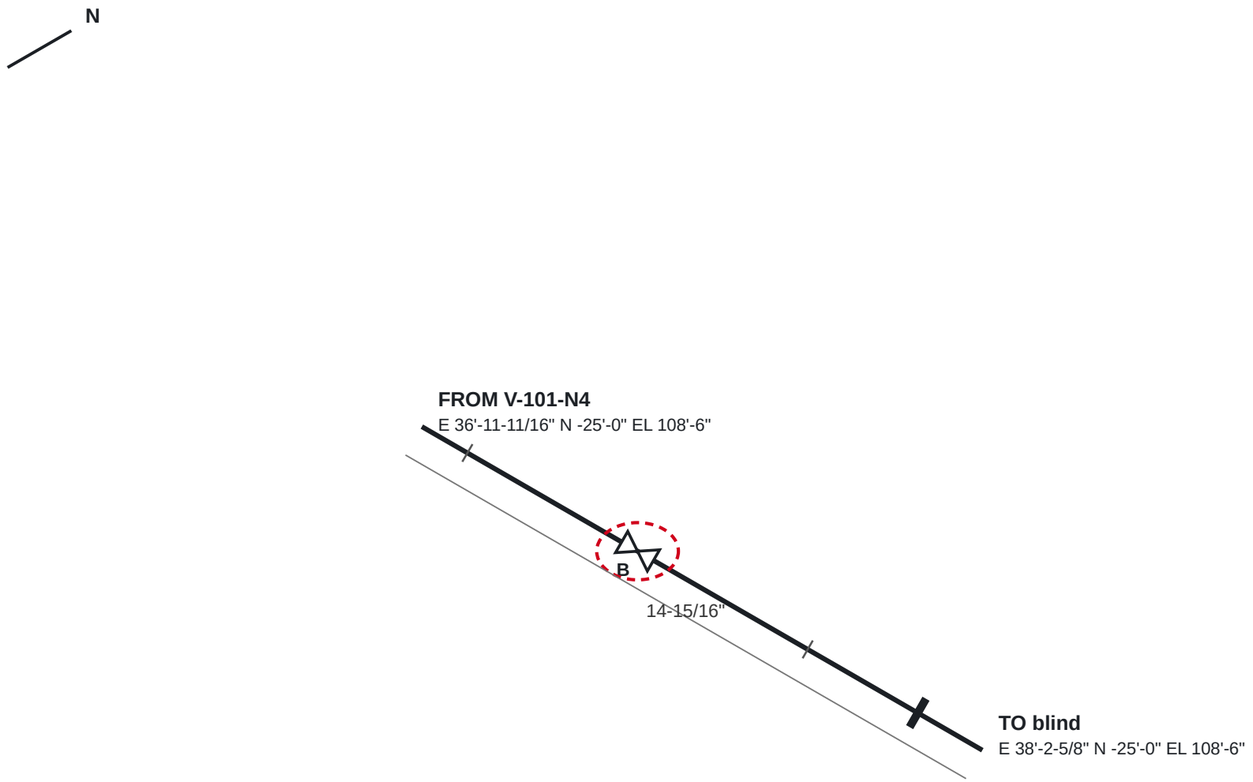
**DWG ISO-1"-P150-V-6001 REV 0 WO-200-02**

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## Isometric ISO-1"-P150-D-6002

ISOMETRIC — 1"-P150-D-6002 — Internal inspection of vertical vessel V-101




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 1" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 1" P150 | 1 |
| 3 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 2 |
| 4 | FLANGE BLIND 1" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 1"-P150-D-6002

CLASS H150 — Hydrocarbon process
SERVICE D · 1" · P150 · DESIGN 150 psi
V-101-N4 → blind
PIPE 0.000" · WEIGHT 39 lb

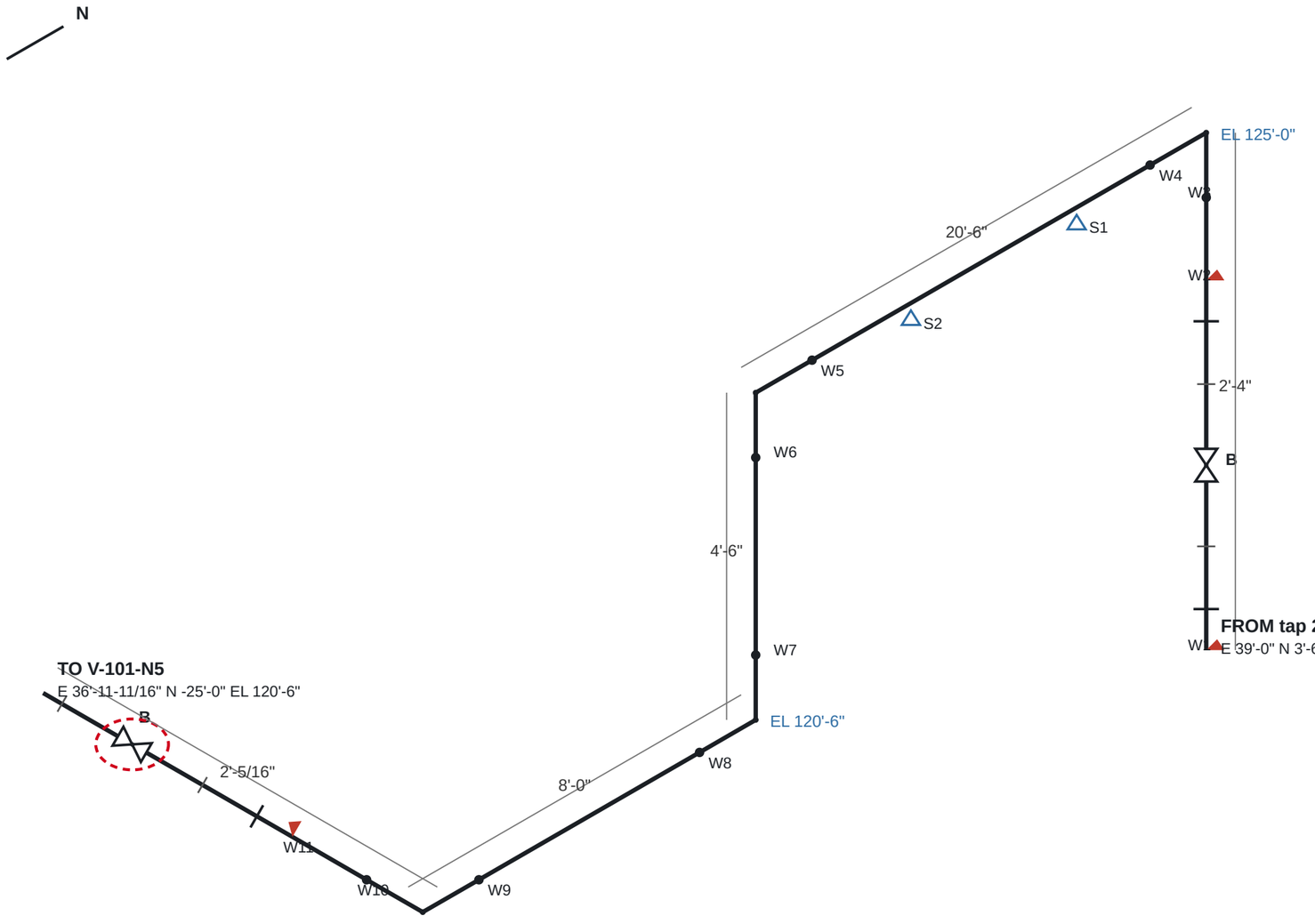
**DWG ISO-1"-P150-D-6002 REV 0 WO-200-02**

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## Isometric ISO-1"-P150-N-3011

ISOMETRIC — 1"-P150-N-3011 — Internal inspection of vertical vessel V-101




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | FLANGE WELD-NECK 1" P150 RAISED FACE | 3 |
| 2 | GASKET 1" P150 COMPRESSED SHEET | 4 |
| 3 | VALVE, BALL, FLOATING 1" P150 | 2 |
| 4 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 4 |
| 5 | PIPE 1" P150 BEVELLED ENDS | 32'-11-3/16" |
| 6 | ELBOW 90° R = 5/3 OD 1" P150 | 4 |

**WELDS 11 (3 FIELD) · SUPPORTS 2 · SPOOLS 4**

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11F

# 1"-P150-N-3011

CLASS G150 — Air and gases
SERVICE N · 1" · P150 · DESIGN 150 psi
tap 2"-P150-N-2007 → V-101-N5
PIPE 32'-11-3/16" · WEIGHT 114 lb

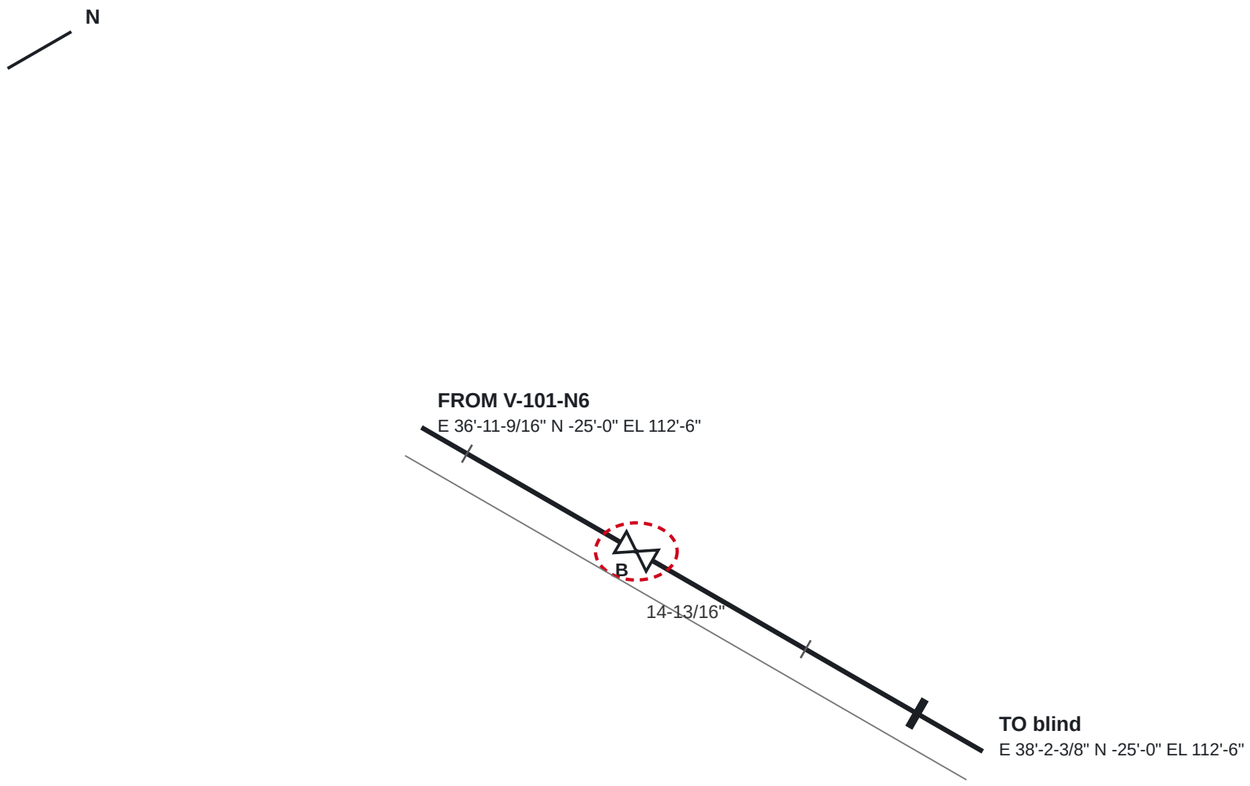
**DWG ISO-1"-P150-N-3011 REV 0 WO-200-02**

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## Isometric ISO-3/4"-P300-I-6003

ISOMETRIC — 3/4"-P300-I-6003 — Internal inspection of vertical vessel V-101




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 3/4" P300 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 3/4" P300 | 1 |
| 3 | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300 | 2 |
| 4 | FLANGE BLIND 3/4" P300 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 3/4"-P300-I-6003

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
V-101-N6 → blind
PIPE 0.000" · WEIGHT 32 lb

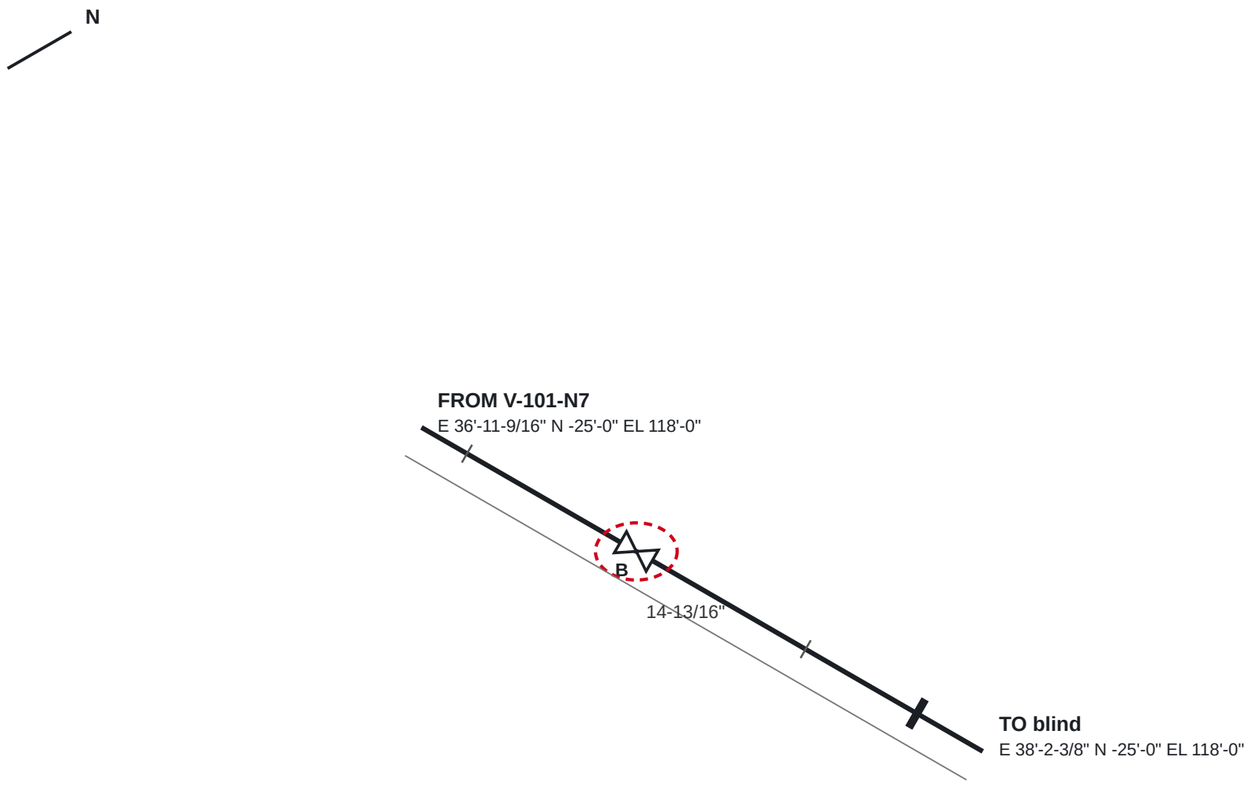
**DWG ISO-3/4"-P300-I-6003 REV 0 WO-200-02**

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## Isometric ISO-3/4"-P300-I-6004

ISOMETRIC — 3/4"-P300-I-6004 — Internal inspection of vertical vessel V-101




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 3/4" P300 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 3/4" P300 | 1 |
| 3 | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300 | 2 |
| 4 | FLANGE BLIND 3/4" P300 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 3/4"-P300-I-6004

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
V-101-N7 → blind
PIPE 0.000" · WEIGHT 32 lb

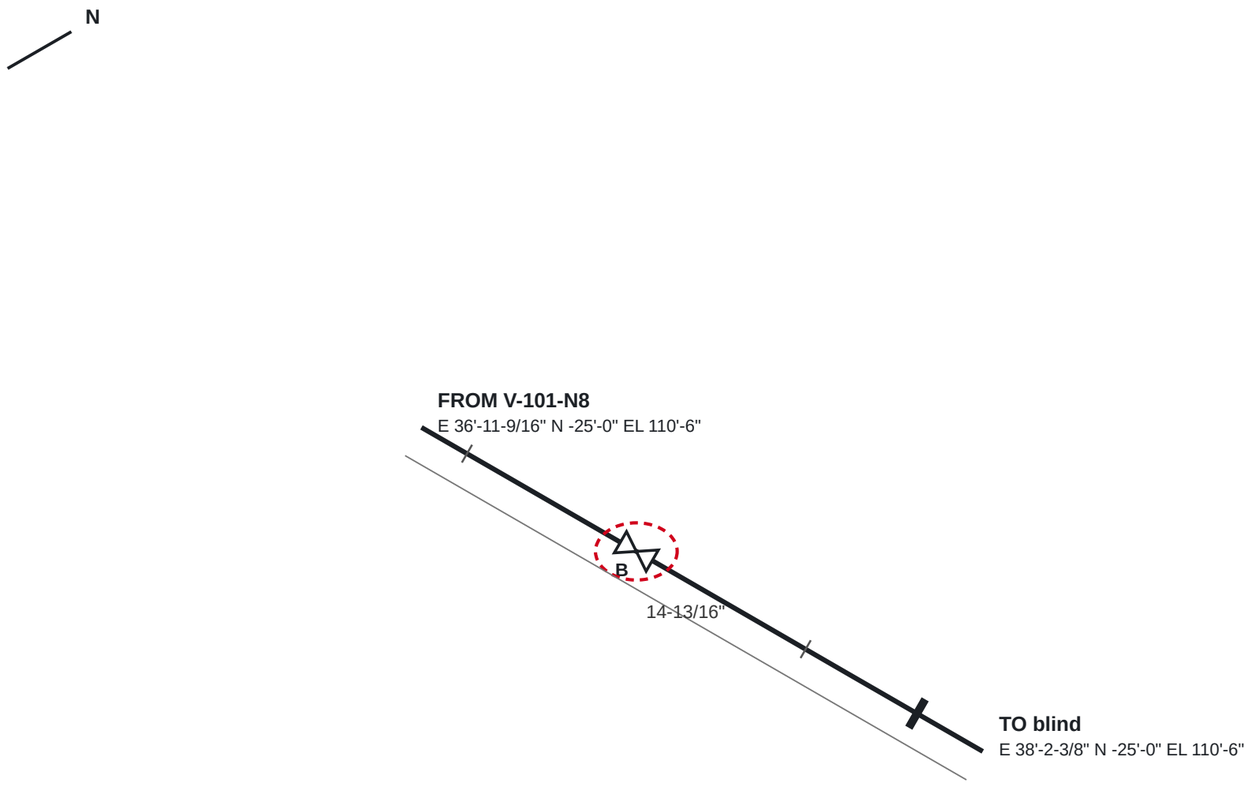
**DWG ISO-3/4"-P300-I-6004 REV 0 WO-200-02**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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## Isometric ISO-3/4"-P300-SP-6005

ISOMETRIC — 3/4"-P300-SP-6005 — Internal inspection of vertical vessel V-101




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 3/4" P300 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 3/4" P300 | 1 |
| 3 | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300 | 2 |
| 4 | FLANGE BLIND 3/4" P300 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 3/4"-P300-SP-6005

CLASS H300 — Hydrocarbon process
SERVICE SP · 0.75" · P300 · DESIGN 300 psi
V-101-N8 → blind
PIPE 0.000" · WEIGHT 32 lb

**DWG ISO-3/4"-P300-SP-6005 REV 0 WO-200-02**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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